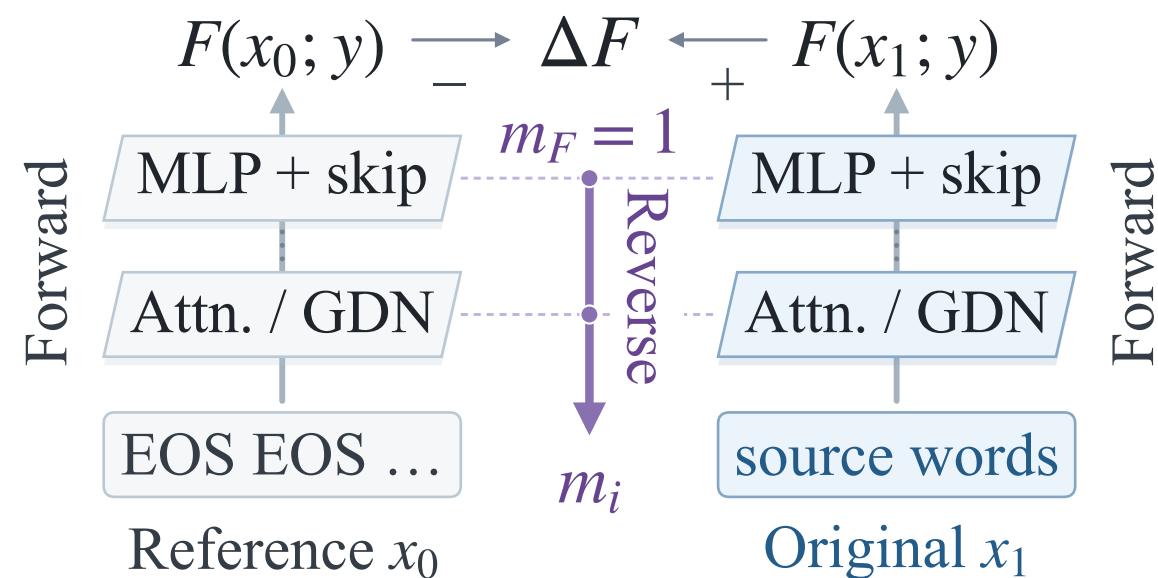


(a) From input change to credit

1 Two forward runs; fixed response y



2 Local rules
from both runs

3 Reverse

propagate coefficients m

4 Token score

$$\Delta h = D_f \Delta a$$

$$m_a = D_f^\top m_h$$

$$A_i = \langle m_i, \Delta e_i \rangle$$

(b) Attention

Select and carry

$$P \times V = Y$$

P V Y

$$\bar{P} = (P_0 + P_1)/2, \quad \bar{V} = (V_0 + V_1)/2$$

Reverse

$$\Delta Y = \boxed{\bar{P} \Delta V} + \boxed{\Delta P \bar{V}}$$

Content Selection

Derive reverse rules

$$M_V = \bar{P}^\top M_Y \quad M_P = M_Y \bar{V}^\top$$

(c) Gated DeltaNet

Store and retrieve



Content

Control

Stored values

Memory use

Reverse

Input token scores

(d) Choose the playwright, not the composer.

Qwen3.5, MoreHopQA example 1

CONTEXT EXCERPTS

... William Shakespeare's play

... The score is by William Walton.

REQUESTED ROLE (EXCERPT)

... the author of the play ...



Name-span total (nats)

+1.56

-1.13

Token contribution (nats)

≤ -2.5 0 $\geq +2.5$

Shown spans underlined

Answer

William
Shakespeare

$$\sum_i A_i = \Delta F$$

Across all input sources